Coding Theory

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To separate the contents of the course to actual additions or out of context information, a black band will be added by its side like the one on this comment.

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Chapter 1

Introduction

Coding theory is part of the information theory, which studies the channel encoding to be able to correct and avoid both erasures and errors when being sent and received through the channel channel.

Example: Consider the set of messages $M = \{1, 0\}$, and suppose that the message "Yes" which will be represented by 1 is sent through a noisy channel. There is a chance that the answer that is received would be 0. To avoid such noise to affect the message drastically, consider the channel encoding that maps $\{0,1\} \to \{0,1\}^3, i \mapsto iii$, then by sending the encoded message, if there is a single bit flip then we can decode by taking the message to be the one that is repeated most.

Definition (Conditional Probability): *If A, B two events, then the conditional probability of A knowing B is defined as*

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Definition (Channel): *A channel is the triple (X, Y, P) where X and Y are non-empty finite sets, called the input and output alphabets and P is a conditional probability function:*

$$P : Y \times X \to [0,1], \quad (y, x) \mapsto P(y|x)$$

For the rest of the course X, Y be finite fields, specifically we take $X = Y = \mathbb{F}_q$.

- ***q*-ary Symmetric Channel:** for $x, y \in \mathbb{F}_q$, $\lambda \in [0,1]$ then

$$P(y|x) = \begin{cases} 1 - \lambda & \text{if } y = x \\ \frac{\lambda}{q-1} & \text{if } y \neq x \end{cases}$$

Note: *We assume that $\lambda < 1/2$ in the case that $q = 2$ given that if $\lambda \geq 1/2$ then by Shannon theorem which states that it is impossible to have a reliable communication with $\lambda \geq 1/2$.*

- **Erasure Channel:** by taking $X = \mathbb{F}_q$ and $Y = \mathbb{F}_q \cup \{\varepsilon\}$ where ε is the empty word, then

$$P(y|x) = \begin{cases} 1 - \lambda & \text{if } y = x \\ \lambda & \text{if } y = \varepsilon \\ 0 & \text{otherwise} \end{cases}$$

Definition (Discrete Memoryless Channel): *Let $n \in \mathbb{N}$, the discrete memoryless channel is $(\mathbb{F}_q^n, \mathbb{F}_q^n, P^n)$ such that*

$$P^n : \mathbb{F}_q^n \times \mathbb{F}_q^n \to [0,1]$$
$$\left(\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \right) \mapsto P^n(y|x) = \prod_{i=1}^{n} P(y_i|x_i)$$

Definition (Error Correcting Code / Code-Words): *An error-correcting code (ECC), $\mathcal{C}$ over the alphabet $\mathbb{F}_q$ of length n is a subset $\mathcal{C} \subset \mathbb{F}_q^n$, for which its elements are called code-words.*

Definition (Hamming Distance/Support/Weight):

- *The Hamming distance is a distance on $\mathbb{F}_q^n$ defined as*

$$d_H : \mathbb{F}_q^n \times \mathbb{F}_q^n \to \mathbb{N}$$
$$\left(\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \right) \mapsto \#(i \in [\![1, n]\!] \mid x_i \neq y_i)$$

- *The support of $x \in \mathbb{F}_q^n$ is defined as $\operatorname{supp}(x) = \{i \in [\![1, n]\!] \mid x_i \neq 0\}$*
- *The weight of $x \in \mathbb{F}_q^n$ is defined as $\operatorname{wt}(x) = \# \operatorname{supp}(x) = d_H(x, \overline{0})$.*

Definition (Minimum Distance): *The minimum distance of $\mathcal{C} \subset \mathbb{F}_q^n$ is defined as $d(\mathcal{C}) := \min\{d(x,y) \mid x,y \in \mathcal{C}, x \neq y\}$.*

Definition ((n,M,d)-Codes): *An (n,M,d)-code is a code $\mathcal{C}$ of length n, minimum distance d and $\# C = M$.*

Example:

1. The repetition code: $\mathcal{C} = \left\{(\alpha)_{i=1}^n \mid \alpha \in \mathbb{F}_q\right\} \subset \mathbb{F}_q^n$, the minimum distance is $d(\mathcal{C}) = n$ and $\# C = q$. This is a (n,q,n)-code.
2. The parity-code: $\mathcal{C} = \left\{(x_1,\dots,x_n) \mid x_n = -\sum_{i=1}^{n-1} x_i\right\} \subset \mathbb{F}_q^n$, $d(\mathcal{C}) = 2$, $\# C = q^{n-1}$, which is a $(n, q^{n-1}, 2)$-code.

Definition: *Let $\mathcal{C} \subset \mathbb{F}_q^n$ be a code, a decoder for $\mathcal{C}$ is a map*

$$D : \mathbb{F}_q^n \to \mathbb{F}_q^n \sqcup \{\perp\}$$
$$x \mapsto D(x) = \begin{cases} x & \text{if } x \in C \\ \perp & \text{otherwise} \end{cases}$$

where we call $\perp$ the failure symbol.

- **Maximal Likelihood Decoder (LD)**:

$$D_m : \mathbb{F}_q^n \to C \sqcup \{\perp\}$$
$$y \mapsto D_m(y) = \begin{cases} x & \text{if } x \in \{x \mid P(y|x) \text{ is maximal}\} \\ \perp & \text{otherwise} \end{cases}$$

- **Nearest-Codeword Decoder (NC)**:

$$D_{N_c} : \mathbb{F}_q^n \to C \sqcup \{\perp\}$$
$$y \mapsto D_{N_c} = \begin{cases} x & \text{if } x \in \{x \mid \forall x' \in C, d_H(x,y) \leq d_H(x',y)\} \\ \perp & \text{otherwise} \end{cases}$$

Theorem: *In q-ary symmetric channel, under the assumption of the uniform distribution of the transmitted code-word message. Then the LD decoder and NC decoder are equivalent.*

1.1. Linear Codes

Definition (Linear Code): *A linear code $\mathcal{C} \subset \mathbb{F}_q^n$ is a vector subspace of $\mathbb{F}_q^n$, its length is n, and its dimension is k.*

Definition: *A generator matrix for $\mathcal{C}$ is a $k \times n$ matrix G whose rows are a basis of $\mathcal{C}$.*

If G is a generator matrix for $\mathcal{C}$, an encoding function on the set of messages $\mathcal{M}$ is $c : M \to \mathbb{F}_q^n, x \mapsto xG$.

Definition (Systematic Form): *A generator matrix G is in systematic form if it is of the form $G = [I_k \mid A]$, A is $k \times (n-k)$, in this case $(x_1,\dots,x_k) \mapsto (x_1,\dots,x_k,x_{k+1},\dots,x_n)$.*

Proposition: *If $\mathcal{C}$ is linear, then $d_H(x,y) = d_H(x-y,0) = \mathrm{wt}(x-y)$ and the minimum distance of $\mathcal{C}$ is the smallest weight of $x \in \mathcal{C} \setminus \{0\}$.*

Notation: *If a code $\mathcal{C}$ is linear, we denote it as an $[n,k,d]$.*

Example:

1. Trivial codes: $\mathbb{F}_q^n$ and any $\mathcal{C}$ with $M = 1$. If $\mathcal{C}$ is linear then $\mathcal{C} = \{0\}$.
2. Repetition codes: $\mathcal{C} = \left\{(a,a,\dots,a) \mid a \in \mathbb{F}_q\right\}$ dim 1, basis $\{(1,\dots,1)\}$.
3. Parity codes: $\mathcal{C} = \left\{(x_1,\dots,x_n) \in \mathbb{F}_q^n \mid x_n = -\sum_{i=1}^{n-1} x_i\right\}$ with dim $n-1$, minimum distance 2, basis $(e_i - e_n)_{i=1}^n$.

Proposition: *A linear code $\mathcal{C}$ with $d(\mathcal{C}) = d$ can detect up to $d-1$ errors, and can correct up to $\lfloor \frac{d-1}{2} \rfloor$ errors in a nearest-codeword decoder.*

Definition (Dot Product/Orthogonal): *$\mathbb{F}_q^n$ is equipped with a symmetric bilinear form $\mathbb{F}_q^n \times \mathbb{F}_q^n \to \mathbb{F}_q$ called the dot product defined as $\cdot : \mathbb{F}_q^n \times$*

$\mathbb{F}_q^n \to \mathbb{F}_q, (x, y) \mapsto x \cdot y = \sum_{i=1}^n x_i y_i$, *we say that* x, y *are orthogonal if and only if* $x \cdot y = 0$.

Proposition:
- $(x + x') \cdot (y + y') = x \cdot y + x \cdot y' + x' \cdot y + x' \cdot y'$.
- $\lambda(x \cdot y) = (\lambda x) \cdot y = x \cdot (\lambda y)$

Definition (Orthogonal Code): *The orthogonal code of* $\mathcal{C}$ *is* $\mathcal{C}^\perp = \{x \in \mathbb{F}_q^n \mid \forall y \in \mathcal{C}, x \cdot y = 0\}$.

Proposition: *Let* $\mathcal{C}$ *be a linear code, then* $\mathcal{C}^\perp$ *is also a linear code.*

Definition (Self-Orthogonal/Self-Dual/Hull): *Let* $\mathcal{C}$ *be a linear code.*
- *if* $\mathcal{C} \subset \mathcal{C}^\perp$ *then* $\mathcal{C}$ *is self-orthogonal.*
- *if* $\mathcal{C} = \mathcal{C}^\perp$ *then* $\mathcal{C}$ *is self-dual.*
- $\mathcal{C} \cap \mathcal{C}^\perp$ *is the hull of* $\mathcal{C}$.

Proposition: $G = [I_k \mid A]$ *is the generator matrix of a* $[n, k, d]$ *code* $\mathcal{C}$. *Then* $\mathcal{C}^\perp$ *has a generator matrix of the form* $[-A^T \mid I_{n-k}]$.

Definition (Parity-Check Matrix): *The generator matrix* H *of* $\mathcal{C}^\perp$ *is called the parity-check matrix of* $\mathcal{C}$.

Corollary: *If* $\mathcal{C}$ *is a* k*-dimensional code, then* $\mathcal{C}^\perp$ *is* $(n-k)$*-dimensional.*

Corollary: $(\mathcal{C}^\perp)^\perp = \mathcal{C}$

Definition (Syndrome): *Let* $H = (h_1 \mid \cdots \mid h_n)$ *be the parity-check matrix of a* k*-dimensional code* $\mathcal{C}$. *We define the syndrome application associated with* H *the map*

$$\sigma : \mathbb{F}_q^n \to \mathbb{F}_q^{n-k} \qquad x \mapsto \sigma(x) = xH^t = \sum_{i=1}^n x_i h_i$$

Proposition: *If* H *is the parity-check matrix of* $\mathcal{C}$, *then* $d(\mathcal{C})$ *is the smallest number of linearly dependent codewords of* H.

We can use the syndromes to do decoding. Suppose that we transmit the word $c \in \mathcal{C}$ and receive $y = c + e$ here e is some error, then by finding e with the smallest hamming weight such that $y - e \in \mathcal{C}$, and deduce y to $y - e$. We start by computing the syndrome $s = \sigma(y)$ and look for the smallest $J \subset [\![1, n]\!]$ such that $\exists \lambda_i \in \mathbb{F}_q^\star, i \in J$ and $s = \sigma(y) = \sum \lambda_i h_i$.. Since $\sigma(c) = 0$ then we have that $\sigma(y) = \sigma(e)$, therefore the closest code-word to y is c. Moreover, if $|J| < d/2$ where d is the minimum distance of the code, we show that c is the unique closest code-word to y. Hence, we decode y to c.

In practice, we create a list of syndromes S and a list of e minimal weight vectors such that $\forall s \in S, \exists e \in E, \sigma(e) = s$. When we receive y, we compute its syndrome, find $s \in S$, then take the corresponding vector e to decode to $c = y - e$.

1.2. Bounds

Theorem (Singleton Bound): *For a* $[n, M, d]$*-code* $\mathcal{C}$, *we always have that* $M \leq q^{n-d+1}$, *in the case where* $\mathcal{C}$ *is linear of dimension* k, $k \leq n - d + 1$.

Proof. Let $\varphi : \mathcal{C} \to \mathbb{F}_q^{n-d+1}$, with its last $d-1$ components deleted. Suppose that φ is not injective, we have two different codewords c_1 and c_2 that only disagree in the last $d-1$ components, that is $d_H(c_1, c_2) \leq d - 1$ which is a contradiction since the minimum distance is d. Thus, φ is injective, hence, $\#\mathcal{C} \leq \# q^{n-d+1}$. If $\mathcal{C}$ is linear, then φ is linear, thus $\dim(\mathcal{C}) = k \leq n - d + 1$. $\square$

The codes that achieve this bound are called Maximum Distance Separable.

Theorem (Asymptotic Gilbert-Varsheimov Bound (GV)): *For any rational number $\kappa \in [0,1]$, there exist a family of linear codes indexed by length $\{C_n\}_{n=1}^{\infty}$ with $\frac{k}{n} = \kappa$ such that the quantity $\delta = \liminf_{n\to\infty} \frac{d(C_n)}{n}$ satisfies $\kappa \geq 1 - h(\delta)$ where h is the entropy function which is in the case $q = 2$ defined as $h(\delta) = \delta \log_2\left(\frac{1}{\delta}\right) + (1-\delta)\log_2\left(\frac{1}{1-\delta}\right)$ and in general as $h(\delta) = \delta \log_q\left(\frac{q-1}{\delta}\right) + (1-\delta)\log_q\left(\frac{1}{1-\delta}\right)$.*

In general, an infinite family of codes C_n such that $\frac{k_n}{n} \to \kappa$ will meet the bound with equality for large n with probability 1.

1.3. Reed-Solomon Codes

Let $k \in \mathbb{N}_{>0}$ and consider the vector space $\mathbb{F}_q[X]_{<k} = \{P \in \mathbb{F}_q[X] \mid \deg P < k\}$, Let $x = \{x_1, ..., x_n\} \subset \mathbb{F}_q$ with $i \neq j \Rightarrow x_i \neq x_j$.

Definition (Reed-Solomon Code): *Consider the evaluation function*

$$\mathrm{ev} : \mathbb{F}_Q[X]_{<k} \to \mathbb{F}_q^n \quad P \mapsto (P(x_1), ..., P(x_n))$$

Then we define the Reed-Solomon code as $\mathrm{RS}_k(x) = \mathrm{ev}\left(\mathbb{F}_q[X]_{<k}\right)$.

Theorem: *Let $k \leq n$, then $\mathrm{RS}_k(x)$ is a $[n, k, n+1-k]$-linear code. In particular, it is an MDS.*

Proof. Linearity follows as RS_k is the image of a vector space through a linear map. The length is clearly n, $\dim \mathbb{F}_q[X]_{<k} = k$ which has basis $\{1, X, ..., X^{k-1}\}$. $\mathrm{ev}(P) = (0, ..., 0) \Rightarrow P(x_1) = \cdots = P(x_n) = 0$, but since $n \geq k$ then necessarily $P = 0$ so ev is injective, thus we have that $\dim_{\mathbb{F}_q} \mathrm{RS}_k(x) = k$. Let $c \in \mathrm{RS}_k(x)$, then $c = \mathrm{ev}(P)$ for some $P \in \mathbb{F}_q[X]_{<k}$, $\mathrm{wt}(c) = \#\{i \mid P(x_i) \neq 0\} = n - \#\{i \mid P(x_i) = 0\} \geq n - (k-1) = n - k + 1$ and by the Singleton bound, we have that $d = n + 1 - k$, so it is an MDS code. $\square$

Proposition: *Reed-Solomon code with generator matrix given by*

$$G = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ x_1 & x_2 & x_3 & \cdots & x_n \\ x_1^2 & x_2^2 & x_3^2 & \cdots & x_n^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_1^{k-1} & x_2^{k-1} & x_3^{k-1} & \dots & x_n^{k-1} \end{pmatrix} \in \mathbb{F}_q^{k\times n}$$

Let X be an algebraic projective geometrically irreducible smooth curve of genus g defined over $\mathbb{F}_q$.

Let $D = \sum_{P\in X} n_P \cdot (P) \in \mathrm{Div}(X)$. The space

$$L(D) = \left\{f \in \mathbb{F}_q(X) \setminus \{0\} \mid \forall P \in X, \mathrm{ord}_P(f) + n_P \geq 0\right\} \cup \{0\}$$

in a finite field, we have $\deg D = \sum_{P\in X} n_P \deg P$. Riemann's inequality states that $l(D) \geq \deg\ P + 1 - g$. Consider $X\left(\mathbb{F}_q\right)$ be the set of rational points of X.

Definition (Algebraic Geometry Code): *Let $\mathcal{P} = \{P_1, ..., P_n\} \subset X\left(\mathbb{F}_q\right)$, let $D \in \mathrm{Div}(X)$, $\mathcal{P} \cap \mathrm{Supp}(D) = \emptyset$ and consider the evaluation*

$$\mathrm{ev} : L(D) \to \mathbb{F}_q^n \quad f \mapsto (f(P_1), ..., f(P_n))$$

Then, the algebraic geometry code $\mathrm{AG}(X, D, \mathcal{P})$ is $\mathrm{ev}(L(D))$.

Theorem: *Let $\deg\ D \leq n$ then $\mathrm{AG}(X, D, \mathcal{P})$ is an $[n, k, d]$ with $k = l(D)$ and $d \geq n - \deg(D)$. We have that $n \leq \#\ X\left(\mathbb{F}_q\right) \leq q + 1 + 2g\sqrt{q}$.*

Theorem (TVZ): *Let $q = p^{2m}$, p prime, for any rate $\kappa \in [0,1]$ there exists a sequence of AG codes whose length goes to infinity and where (κ, δ) satisfy $\kappa + \delta \geq 1 - \frac{1}{p^m - 1}$ with $\kappa = \frac{k}{n}, \delta = \frac{d}{n}$.*

Example:

- Consider $\mathbb{F}_4 = \mathbb{F}_2[X]/((X^2+X+1)) = \{0, 1, \alpha, \alpha+1\}$, $k = 2$ then we have $\mathbb{F}_q[X]_{<2} = \{\beta_0 + \beta_1 X \mid \beta_0, \beta_1 \in \mathbb{F}_4\} = \langle 1, X, \rangle$, $\mathrm{ev}(1) = (1,1,1,1)$, $\mathrm{ev}(X) = (0, 1, \alpha, \alpha+1)$, which would give a generator matrix

$$G = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & \alpha & \alpha+1 \end{pmatrix}.$$

- Let $E : y^2 = x^3 + 3x$ over $\mathbb{F}_5$, $\# E(\mathbb{F}_5) = 10$, let $D = 5(\infty) \in \mathrm{Div}(E)$. Let $L(D) = \left\{ f \in \mathbb{F}_4(E) \mid \mathrm{ord}_p(f) \geq 0, \forall P \neq \infty, \mathrm{ord}_p(f) \geq -5 \right\} = \langle 1, x, x^2, y, xy \rangle$. Take $\mathcal{P} = E(\mathbb{F}_5) \setminus \{\infty\}$